

Introduction

In Chapter 3, you used the Datasets library for the first time. During model fine-tuning, you saw three main steps:

- Loading a dataset from the Hugging Face Hub
- Preprocessing the data using `Dataset.map()`
- Loading metrics and computing performance

However, this only covers a small portion of what the Datasets library is capable of.

What You Will Learn in This Chapter

In this chapter, you will take a deeper look at the Datasets library and explore more advanced functionality. You will find answers to several important questions:

1) What if your dataset is not on the Hub?

Not all datasets are available on the Hugging Face Hub. You will learn how to:

- Load datasets from local files
- Work with CSV, JSON, and other data formats
- Import data from external sources

2) How can you slice and manipulate a dataset?

You will learn how to:

- Use specific subsets of a dataset
- Create train/validation splits
- Filter, select, and shuffle data
- Convert datasets to Pandas if necessary

3) What if your dataset is very large?

If your dataset is so large that it exceeds your laptop's RAM capacity, what should you do?

This chapter explains techniques for efficiently handling large datasets without exhausting system memory.

4) What are “Memory Mapping” and “Apache Arrow”?

To improve performance and efficiency, the Datasets library relies on:

- Memory mapping
- Apache Arrow

You will learn how these technologies work and why they are important for scalable data processing.

5) How do you create your own dataset and upload it to the Hub?

You will also learn how to:

- Create a custom dataset
- Properly structure it
- Publish it to the Hugging Face Hub

This chapter builds the foundation for more advanced tokenization and fine-tuning tasks later in the course.